

MECHATRONIC SYSTEM INTEGRATION

MCTA 3203

WEEK 2 :

DIGITAL LOGIC SYSTEM

SECTION 1

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INSTRUCTOR:

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Abstract

This activity focuses on developing a digital number display mainly utilizing a seven segment display and an Arduino Mega microcontroller chip. The main objective is to design and make a simple, effective and working circuit and code to display numbers from 0 to 9 in order at a press of a button. The system is programmed using Arduino IDE, where the microcontroller sends output signals to the seven-segment display through digital pins to control each segment. After testing, we have achieved the desired output of increasing the displayed digit by one with each press of a button. We also added another button to reset the count. This project reflects the modern digital clock and its inner workings, implementing digital electronics.

Introduction

This experiment aims to display a numerical value from 0 to 9 using a 7 segment display as the output. This experiment will show the working principles of simple logic circuit and coding from arduino uno, and implementing the basic binary coded decimal in our programmes.

A 7 segment display is a widely used output to show the numerical value in mechatronics projects. Each segment can be individually controlled to form a number. Additionally the interface also uses current limiting resistors, input output pin configuration and time delay signal in Arduino Uno.

This lab expects that the arduino would successfully display all numerical values at the 7 segment display. When the circuit is correctly wired and programmed, by pushing the pushbutton, the value would increase and be displayed at the 7 segment display. This outcome will verify correct interfacing between the Arduino, the 7-segment display, and the pushbuttons,demonstrating effective application of digital logic and circuit control principles.

Materials and Equipment

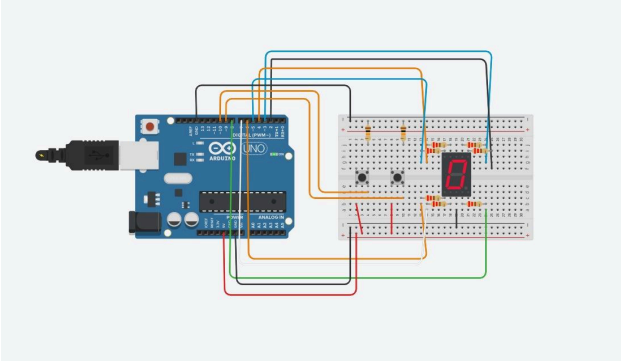
1. Arduino Uno board (1)
2. Common cathode 7-segment display (1)
3. Jumper wires
4. Breadboard (1)
5. Pushbutton (2)
6. 220-ohm resistors (7)
7. 10k-ohm resistors (2)

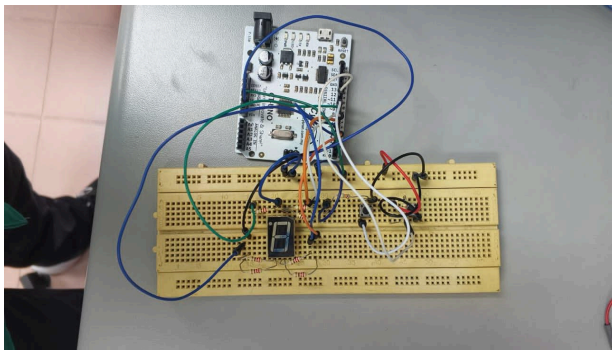
Experimental setup

Circuit Setup:

1. Connect the common cathode 7-segment display to the Arduino Uno as follows:
    - Connect each of the 7 segments (a, b, c, d, e, f, g) of the display to separate digital pins on the Arduino (e.g., D0 to D6).
    - Connect the common cathode pin of the display to one of the GND (ground) pins on the Arduino.
    - Use 220-ohm resistors to connect each of the segment pins to the Arduino pins to limit the current.
2. Connect the pushbuttons to the Arduino:
    - Connect one leg of each pushbutton to a separate digital pin (e.g., D9 and D10) and connect the other leg of each pushbutton to GND.
    - Use 10K-ohm pull-up resistors for each pushbutton by connecting one end of each resistor to the digital pin and the other end to the 5V output of the Arduino.

Circuit diagram





Methodology

Experiment steps:

1. Build the circuit according to the circuit setup instructions.
2. Upload the provided Arduino code to our Arduino Uno.
3. Open the Serial Monitor in the Arduino IDE.
4. Press the increment button to increase the count. The 7-segment display should show the numbers from 0 to 9 sequentially.
5. Press the reset button to reset the count to 0.

IDE code:

```
const int segmentA = 3;
const int segmentB = 2;
const int segmentC = 8;
const int segmentD = 7;
const int segmentE = 6;
const int segmentF = 4;
const int segmentG = 5;

const int button_1 = 9;
const int button_2 = 10;

int count = 0;

int num_array[10][7] = {
  {1, 1, 1, 1, 1, 1, 0},//0
  {0, 1, 1, 0, 0, 0, 0},//1
  {1, 1, 0, 1, 1, 0, 1},//2
  {1, 1, 1, 1, 0, 0, 1},//3
  {0, 1, 1, 0, 0, 1, 1},//4
  {1, 0, 1, 1, 0, 1, 1},//5
  {1, 0, 1, 1, 1, 1, 1},//6
  {1, 1, 1, 0, 0, 0, 0},//7
  {1, 1, 1, 1, 1, 1, 1},//8
  {1, 1, 1, 1, 0, 1, 1}//9
```

```
};

void setup() {
  pinMode(segmentA, OUTPUT);
  pinMode(segmentB, OUTPUT);
  pinMode(segmentC, OUTPUT);
  pinMode(segmentD, OUTPUT);
  pinMode(segmentE, OUTPUT);
  pinMode(segmentF, OUTPUT);
  pinMode(segmentG, OUTPUT);

  pinMode(button_1, INPUT);
  pinMode(button_2, INPUT);

  display(num_array[count]);
}

void loop() {
  if (digitalRead(button_1) == HIGH) {
    if (count < 9) {
      count++;
    }
    else {
      count = 0;//number 0
    }

    display(num_array[count]);

    while (digitalRead(button_1) == HIGH) {
      delay(10);
    }
    delay(50);
  }
```

```
  else if (digitalRead(button_2) == HIGH) {
    count = 0;
    display(num_array[count]);
    while (digitalRead(button_2) == HIGH) {
      delay(10);
    }
    delay(50);
  }
}

void display(int bin[]) {
  digitalWrite(segmentA, bin[0]);
  digitalWrite(segmentB, bin[1]);
  digitalWrite(segmentC, bin[2]);
  digitalWrite(segmentD, bin[3]);
  digitalWrite(segmentE, bin[4]);
  digitalWrite(segmentF, bin[5]);
  digitalWrite(segmentG, bin[6]);
}
```

Data collection

The result was made clearly by observation through our own eyes.

Bil	**Input**	**Observed Output**
1	The arduino is turned on	The 7-segment displays ‘0’
2	The pushbutton_1 is pressed	The 7-segment displays value increment to ‘1’
3	The pushbutton_1 is pressed	The 7-segment displays value increment to ‘2’
4	The pushbutton_1 is pressed	The 7-segment displays value increment to ‘3’
5	The pushbutton_1 is pressed	The 7-segment displays value increment to ‘4’

6	The pushbutton_1 is pressed	The 7-segment displays value increment to ‘5’
7	The pushbutton_1 is pressed	The 7-segment displays value increment to ‘6’
8	The pushbutton_1 is pressed	The 7-segment displays value increment to ‘7’
9	The pushbutton_1 is pressed	The 7-segment displays value increment to ‘8’
10	The pushbutton_1 is pressed	The 7-segment displays value increment to ‘9’
11	The pushbutton_1 is pressed	The 7-segment displays value increment to ‘0’
12	The pushbutton_2 is pressed	The 7-segment displays ‘0’
13	The pushbutton_2 is pressed when the 7-segment displays ‘5’	The 7-segment displays ‘0’

Data Analysis

The data confirmed that:

1. The pushbutton_1 is correctly set to the increment function.
2. The pushbutton_2 is correctly set to reset the current value to zero.
3. The display followed the correct segment configuration defined in the code.

Digit	Active Segment	Observed Display
0	a,b,c,d,e,f	Correct
1	b,c	Correct
2	a,b,d,e,g	Correct
3	a,b,c,d,g	Correct
4	b,c,f,g	Correct
5	a,c,d,f,g	Correct
6	a,c,d,e,f,g	Correct

7	a,b,c	Correct
8	a,b,c,d,e,f,g	Correct
9	a,b,c,d,f,g	Correct

Result

The experiment successfully shows manual control of 7-segment display using Arduino Uno and push buttons.

1. The 7-segment displays the correct value of the current value stored in the Arduino Uno.
2. The value increases by one when the pushbutton_1 is pressed.
3. The value then successfully reset to zero when the second push button is pressed.
4. The segments of the display are correctly connected to the Arduino code logic.
5. The display worked correctly without error or flickering.

Discussion

1. Interpretation of Results and Implications

   The output achieved from the 7-segments display shows that the circuit is well connected to the Arduino Uno. It also shows the code used is suitable for the experiment. In addition, the push buttons used to increment and reset the value also shows integration of input and output shows the result as intended.

2. Discrepancies Between Expected and Observed Outcomes

   There were no major discrepancies observed during the experiment and everything went as intended.

3. Sources of Error and Improvement
   a. Mechanical Button Bounce: Push buttons naturally produce multiple electrical signals once pushed, producing multiple signals. By using a debouncer hardware, this situation can be avoided.
   b. Human Reaction Time: Since the reset and increment push button is manually pressed, the pressing speed and duration may vary resulting in inconsistent results or missed counts.

Conclusion

This project successfully demonstrated the design and implementation of a seven-segment display system using an Arduino microcontroller. The system was able to display numerical digits from 0 to 9 accurately by controlling the output signals sent to each segment. After going through the process of putting the circuit together, coding it and testing, the project achieved its main objective that is - understand and implement digital logic interfacing between Arduino and external display devices, 7-segment display.

In terms of understanding, we now know how to code a counting machine and as for implementing, we are confident in the reenactability of this project. The results proved that even simple electronic systems can be effectively developed through systematic design and logical programming.

For an Islamic reflection, this project highlights the importance of seeking knowledge (Ilm') seen in our objective to understand its inner workings. Knowledge is a gift from Allah that enables mankind to understand and control the world around them. By exploring the principles of electronics and control systems, we recognize the precision and order that Allah has placed in creation.

Recommendations

1. Improve Display Range

   Extend the display range to include two 7-segment displays to show two-digit numbers (00-99). This can be achieved by using a multiplexer or by connection and additional display with separate control pin.

2. Enhance User Interface

   Add more push buttons for more additional functions such as decrement, pause, or auto-count mode to make the system more interactive.

3. Add Light or Sound Indicators

   Connect a buzzer or an LED to indicate a button is pressed or when the counter exceeds certain limits or when it resets to zero for better user feedback.

4. Decimal Value

   With a potentiometer and two 7-segment display connected to show decimal points value to experiment with coding to change analog value received from the potentiometer to a digital output.

References

- In depth guide on Seven Segment Display
(https://lastminuteengineers.com/seven-segment-arduino-tutorial)

Acknowledgments

Alhamdulillah, all praise and gratitude be to Allah SWT for His blessings, guidance, and mercy that enabled us to successfully complete this project entitled “Digital Logic System.” Throughout the development process, we faced several challenges, but with His will and guidance, we managed to overcome them and complete the project successfully.
We would also like to express our heartfelt appreciation to our lecturer for their continuous guidance, support, and valuable feedback throughout this project. Their advice has greatly helped us in understanding both the theoretical and practical aspects of microcontroller systems and digital display design.

Student’s Declaration

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or persons.

We hereby certify that this report has not been done by only one individual and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have read and understand the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for marking and this final printed report has been verified by us.

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